

PROFESSIONAL SUMMARY

Software Engineer with a strong foundation in building scalable, enterprise-grade backend systems (**.NET 10, ASP.NET Core, Clean Architecture**) and responsive fullstack web applications (**Nuxt 4, React, React Native**). Experienced in processing geospatial data (**PostGIS**), AI-driven multimodal evaluations (**Gemini AI, pgvector**), and robust RESTful API design. Passionate about software craftsmanship, clean code, and type-safe systems that solve real-world operational problems.

TECHNICAL SKILLS

Languages: C# (13), TypeScript, JavaScript, SQL, Dart, HTML5/CSS3
Backend & Systems: ASP.NET Core (.NET 10), Clean Architecture, EF Core, Node.js, Express.js, SignalR, RESTful APIs, JWT
Databases & Storage: PostgreSQL (PostGIS spatial queries, pgvector semantic search), MongoDB, Redis
Frontend & Mobile: Nuxt 4 / Vue 3, React.js, React Native, Astro 5, Tailwind CSS, Pinia, Webpack / Vite
DevOps & Tooling: Docker, Docker Compose, Git & GitHub Actions (CI/CD), Linux, Nginx, QuestPDF, Postman, OpenSpec

WORK EXPERIENCE

Geospatial Data Processing · [Aero Asahi Viet Nam](#)

2025 — 2026 | Project: Oruso

- Processed, cleaned, and analyzed large-scale spatial datasets to meet stringent accuracy standards for Japanese GIS project specifications.
- Executed spatial validation pipelines to identify topology anomalies, attribute discrepancies, and coordinate alignment errors.

Frontend Developer Intern · [UTA Solution](#)

2024 — 2025 | Project: Pago

- Developed responsive user interfaces for agricultural production management web platform using ReactJS and Tailwind CSS.
- Implemented interactive data visualization widgets for harvest analytics and assisted in evaluating computer vision AI models for early pest detection.

FEATURED PROJECTS

TechDaily — Developer Micro-Learning & AI Interview Platform

2026 | github.com/DuyclD03/techdaily

Tech stack: .NET 10, C# 13, Clean Architecture, Nuxt 4, PostgreSQL (pgvector), Gemini AI, Quartz.NET

- Architected fullstack micro-learning platform using Clean Architecture with pure dependency injection (no third-party mediator overhead).
- Engineered semantic search and term explainer utilizing PostgreSQL **pgvector** cosine similarity indexing.
- Integrated Google Gemini 2.5 Flash for 1-pass multimodal voice & text evaluation with strict structured JSON output schemas.

MotelLease — Boarding House & Property Management System

2024 — Present | github.com/DuyclD03/motellease

Tech stack: ASP.NET Core (.NET 10), PostGIS, Nuxt 4, EF Core, SignalR, QuestPDF, MoMo & VNPay, Docker

- Engineered enterprise boarding house management serving 4 distinct user roles (Tenant, Landlord, Staff, Admin) with spatial map search using **PostGIS GiST** indexes.
- Implemented automated monthly utility billing with vector PDF generation (**QuestPDF**) and realtime WebSocket alerts via **SignalR**.
- Built idempotent payment processing (MoMo, VNPay HMAC-SHA512) and verified business invariants across 140+ integration tests.

Boarding House Booking — Capstone Graduation Project (Grade: 8+/10) 2024 | github.com/DuyclD03/Boarding-house-booking

Tech stack: React.js, React Native, Node.js, Express.js, MongoDB, Cloudinary, VNPay Gateway

- Delivered complete accommodation discovery and booking system across Web application, Admin dashboard, and Mobile app (iOS/Android).
- Designed RESTful APIs with JWT authentication, role-based authorization, and automated rent calculation with VNPay integration.

EDUCATION

FPT University

Bachelor of Software Engineering · GPA: 8.3 / 10

2021 — 2025

CERTIFICATIONS

- Software Development Lifecycle (Coursera)
- Computer Communications (Coursera)
- Web Design for Everybody (Coursera)